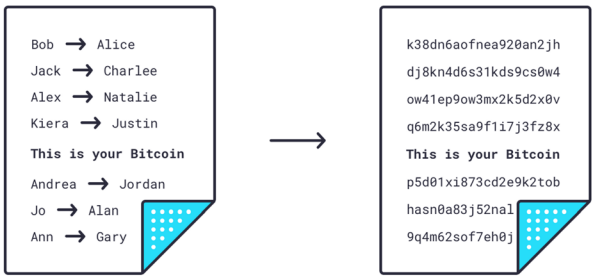
- Just as a slight deviation in a musical performance can cause it to sound different, even the slightest change to the original data will result in a different hash value. This makes hashing a powerful tool for ensuring the integrity and authenticity of digital information.

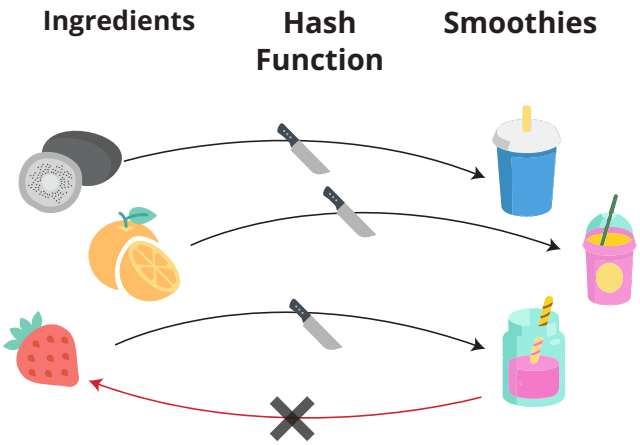
The process of encoding the **public key** through hashing is used to improve the security of information by converting it into a fixed-length, unreadable format. Bitcoin uses the SHA-256 and Ripemd-160 algorithms to produce public addresses. The resulting output serves as a unique identifier for the **public key** and helps to ensure the integrity and security of transactions stored in the *blockchain*. By **encoding** the information in this way, it becomes more difficult for unauthorized individuals to access and manipulate the data.



In this link you'll find another hash generator where you can try it out yourself.

Hashing

A hash function takes any input, and produces a fixed-length output (hash).



- ***Deterministic.*** The same ingredients always yield the same smoothie.

- ***Pre-Image Resistance.*** You can't glue together a strawberry when given a smoothie.

- ***Correlation Resistance.*** Changing the ingredients a little results in a completely different smoothie.

- ***Collision Resistance.*** It's hard to find different ingredients for a smoothie that result in the exact same one.

- ***Speed & Verifiablility.*** Throw fruit into the mixer. It's fast and what comes out for sure is a smoothie.